

This report examines daily returns and liquidity for 23 cryptocurrencies using daily open, high, low, close, volume and market capitalisation data. The combined dataset contains 37,082 observations and is an unbalanced panel, as some cryptocurrencies have longer trading histories than others. Daily log returns are calculated as:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right)$$

where $P_{i,t}$ is the closing price of cryptocurrency i on day t . The first return observation for each cryptocurrency is missing because a previous closing price is required.

Figure 1: Daily log return plots

The selected plots show Bitcoin, Dogecoin, Solana and Tether to illustrate variation across a large benchmark cryptocurrency, a speculative cryptocurrency, a newer high-volatility cryptocurrency and a stablecoin.

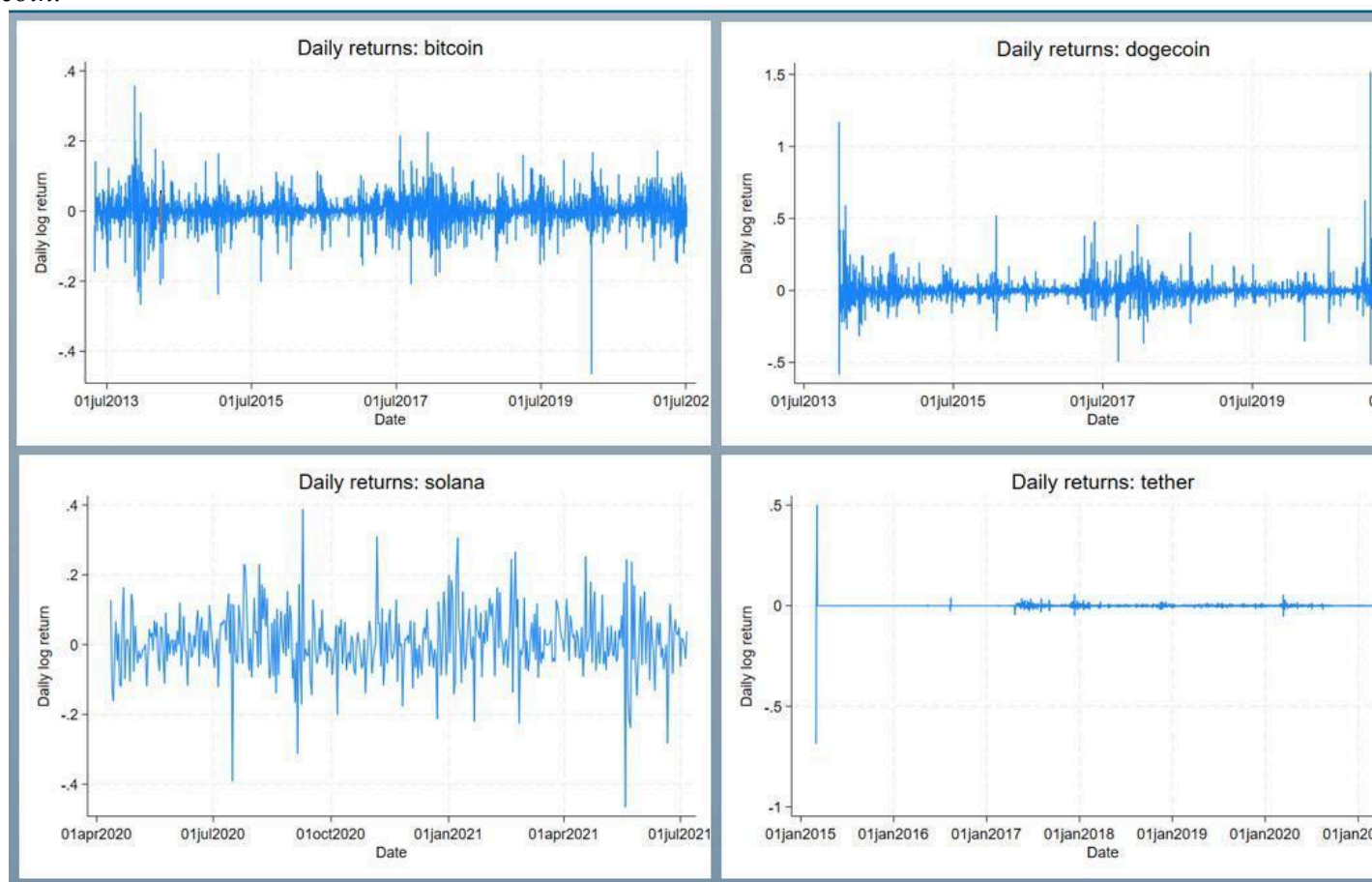


Figure 1 shows substantial volatility in daily cryptocurrency returns. Across the full sample, the mean daily log return is 0.0021, meaning the average daily return is slightly positive. However, the standard deviation is much larger at 0.0684, showing that daily returns vary widely around the mean. The minimum return is -0.6871 and the maximum is 1.5164, indicating extreme downside and upside movements. This supports the visual evidence that cryptocurrency returns are volatile and affected by large outliers.

The four selected coins show contrasting cryptocurrency behaviour, rather than representing the whole sample equally. Bitcoin is included as a large benchmark cryptocurrency but still displays sharp daily movements. Dogecoin shows the clearest speculative behaviour with very large return spikes. Solana has a shorter history and the highest mean return among the reported coin averages, at 0.0084, but also visibly high volatility. Tether behaves differently because it is a stablecoin: its mean return is close to zero and its returns are mostly flat, except for unusual early observations. Overall, the selected plots support coin-level analysis rather than treating all cryptocurrencies as one identical asset type.

Liquidity is measured primarily using the Amihud illiquidity measure:

$$Amihud_{i,t} = \frac{|r_{i,t}|}{DollarVolume_{i,t}}$$

where:

$$DollarVolume_{i,t} = Close_{i,t} \times Volume_{i,t}$$

Amihud captures price movement relative to trading value. A higher value indicates greater illiquidity, because trading is associated with a larger price movement. This follows Amihud's (2002) price-impact approach to illiquidity.

In addition to Amihud, I selected turnover and the high-low range as additional daily-data liquidity proxies, because the dataset contains daily OHLC prices, volume and market capitalisation, but not bid-ask spreads, order-book depth, intraday trades or transaction costs. Turnover is calculated as:

$$Turnover_{i,t} = \frac{Volume_{i,t}}{MarketCapitalisation_{i,t}}$$

Turnover captures trading activity relative to market size and has been used as a liquidity proxy in previous research (Datar, Naik and Radcliffe, 1998). Higher turnover may suggest greater liquidity, although in cryptocurrency markets it may also reflect speculative attention or market stress.

The high-low range is calculated as:

$$HighLowRange_{i,t} = \frac{High_{i,t} - Low_{i,t}}{Close_{i,t}}$$

This captures daily price movement relative to the closing price. It is not treated as a direct bid-ask spread measure, but high and low prices have been used to infer liquidity-related trading costs from daily data (Corwin and Schultz, 2012). Here, it is interpreted cautiously as a price-range proxy that may reflect volatility, price instability or trading frictions.

Figure 2: Amihud illiquidity plots

The selected plots show Bitcoin, NEM, Dogecoin and Tether to illustrate variation across a liquid

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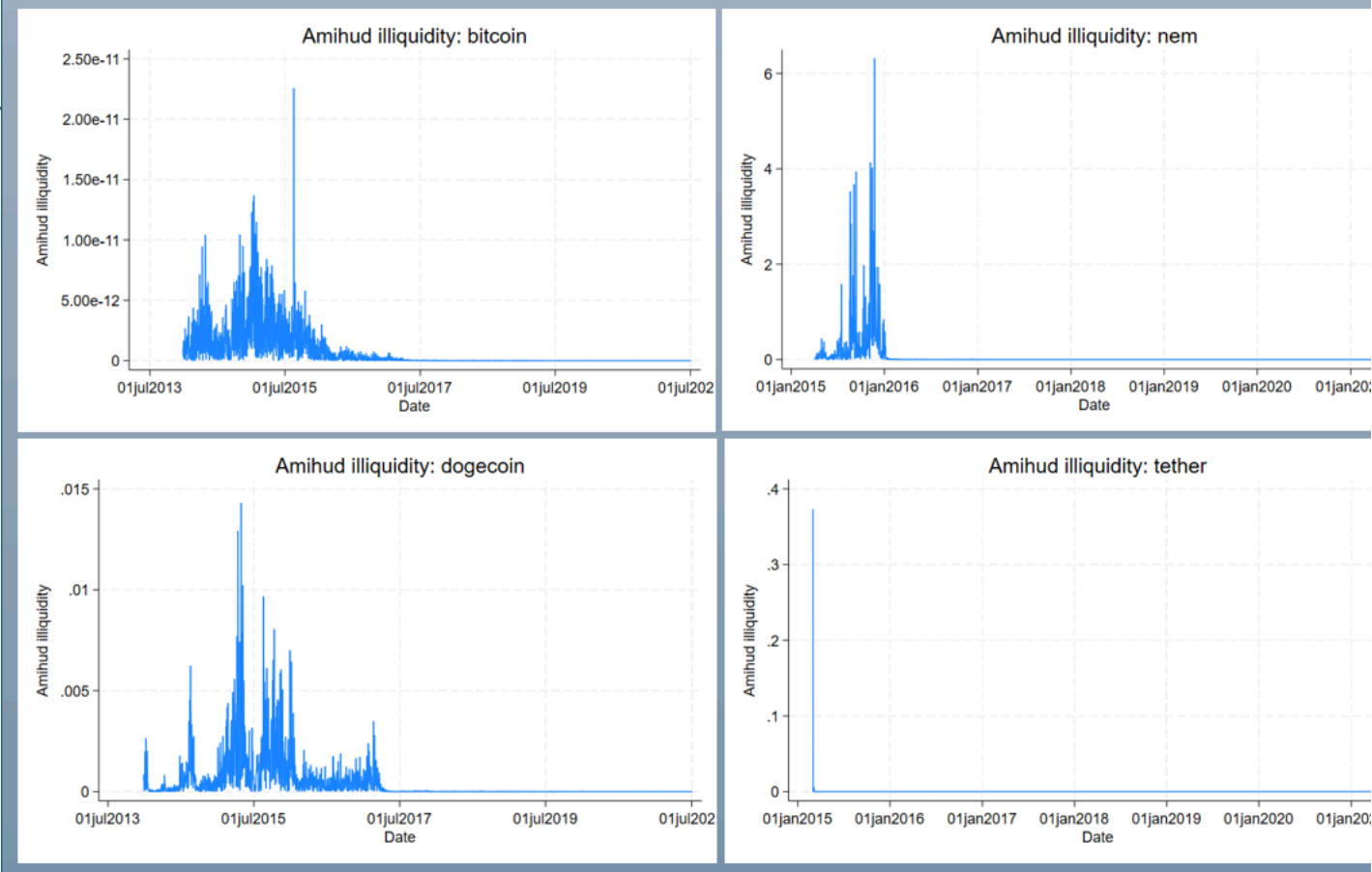


Figure 2 shows large differences in Amihud illiquidity across the selected cryptocurrencies. Bitcoin has the lowest mean Amihud in the sample, at 6.75×10^{-13} , consistent with its role as a highly traded benchmark. By contrast, NEM has the highest mean Amihud, at 0.0537, with extreme illiquidity spikes. Dogecoin also shows clustered spikes, with a mean Amihud of 0.0003977, suggesting periods when price movements were large relative to trading value. Tether differs again: its mean Amihud is 0.0001637, but the plot is dominated by an early spike, after which the series appears almost flat. This reflects stablecoin behaviour and Amihud's sensitivity to unusual daily observations. It also highlights a measurement issue: applying the same liquidity measure across speculative cryptocurrencies and stablecoins may produce different interpretations.

Table 1: Descriptive summary by cryptocurrency

Cryptocurrency	an daily return	Mean Amihud	Mean turnove	Mean high-low range
Bitcoin	0.0018	6.75×10^{-13}	0.0541	0.0500
Dogecoin	0.0026	0.0003977	0.1163	0.0861
Solana	0.0084	7.65×10^{-9}	0.1310	0.1368
Tether	-0.0001	0.0001637	2.2595	0.0111
NEM	0.0026	0.0537	0.0318	0.1039

Table 1 supports the figure-based evidence and reports the additional daily-data liquidity proxies alongside mean returns and mean Amihud. Bitcoin has the lowest average Amihud, while NEM has the highest, matching Figure 2. Solana has the highest mean return among the selected coins, at 0.0084, and the highest mean high-low range, at 0.1368, suggesting that higher returns are accompanied by greater daily price movement. Tether has a near-zero mean return and the lowest high-low range, consistent with stablecoin behaviour.

To test for an illiquidity premium, I estimate three regressions of daily returns on lagged liquidity variables. Lagging helps test whether yesterday's liquidity predicts today's return, rather than same-day co-movement. Standard errors are adjusted for heteroskedasticity throughout. Turnover and high-low range are included in

Model 3 as additional daily-data proxies alongside Amihud.

Table 2: Liquidity premium regression results

Variable	Model 1	Model 2	Model 3
Lagged log Amihud	0.000128*** (0.0000468)	0.000298*** (0.0000848)	0.001002*** (0.0001534)
Lagged log turnover	—	—	0.003328*** (0.0004989)
Lagged high-low range	—	—	0.022991* (0.013068)
Cryptocurrency fixed effects	No	Yes	Yes
Observations	35,801	35,801	35,478
R-squared	0.0002	0.0010	0.0059

Table 2 tests whether lagged illiquidity predicts next-day cryptocurrency returns. In Model 1, lagged log Amihud is positive and statistically significant, with a coefficient of 0.000128 and p-value of 0.006. This suggests higher illiquidity is associated with higher subsequent returns, consistent with an illiquidity premium. However, the R-squared is only 0.0002, so Amihud alone explains very little daily return variation.

Model 2 adds cryptocurrency fixed effects to control for persistent differences between coins, such as size, maturity and average trading activity. The Amihud coefficient remains positive and increases to 0.000298, with a p-value below 0.001. This suggests the relationship is not driven only by differences between coins, but also appears within cryptocurrencies over time.

Model 3 adds turnover and the high-low range as additional daily-data proxies. Lagged log Amihud remains positive and significant, rising to 0.001002, so the result still holds after these controls. Lagged log turnover is also positive and significant, at 0.003328, although this may capture trading attention, speculation or market stress rather than liquidity alone. The high-low range is positive but only weakly significant, with a

p-value of 0.079. Although the R-squared rises to 0.0059, explanatory power remains low. Overall, the results are consistent with an illiquidity premium, but do not prove that liquidity is the main driver of

cryptocurrency returns. Other factors, including sentiment, volatility, news and coin-specific events, are likely to explain much of the remaining variation.

Part 2:

Estimating liquidity with daily data is difficult because the available variables only provide a low-frequency summary of trading conditions. Table 1 shows that turnover and the high-low range can be calculated from daily volume, market capitalisation and OHLC prices, but these variables do not reveal bid-ask spreads, order-book depth, transaction costs or intraday price impact. This matters because an asset can have high daily volume but still be costly to trade in size if large orders move prices. Similarly, the high-low range captures daily price movement, but cannot fully separate liquidity frictions from volatility. This is important because Corwin and Schultz (2012) show that high and low prices can be used to infer liquidity-related trading costs, but they remain indirect estimates. Therefore, the measures in Table 1 are useful daily proxies rather than precise measures of market liquidity.

These limitations are particularly relevant for cryptocurrencies. Crypto markets trade continuously, but daily data imposes artificial daily snapshots using closing prices. This can hide intraday liquidity shocks, temporary exchange disruptions or short periods of thin trading. Figure 1 shows large return outliers for Dogecoin and Solana, which matters because Amihud uses absolute returns in the numerator. As a result, unusually large returns can mechanically increase measured illiquidity, even when the price movement may partly reflect speculation rather than trading difficulty. Figure 2 shows this issue through extreme Amihud spikes, especially for NEM and Tether. Tether is particularly important because Table 1 shows a near-zero mean return and low high-low range, while Figure 2 shows an Amihud series dominated by an early outlier. This suggests that applying the same daily liquidity measure across speculative cryptocurrencies and stablecoins can produce different interpretations.

Table 2 is therefore informative but should be read cautiously. Lagged Amihud is positive and statistically significant in all three models, which is consistent with an illiquidity premium. However, the low R-squared values show that liquidity explains only a small proportion of daily return variation. This means the result supports the idea that illiquidity is priced, but it does not show that liquidity is the main driver of cryptocurrency returns.

The findings are broadly consistent with asset-pricing research showing that liquidity can be priced. Amihud (2002) finds that expected stock returns increase with expected illiquidity, supporting the idea that investors require compensation for holding less liquid assets. Pástor and Stambaugh (2003) show that exposure to

market-wide liquidity risk is related to expected stock returns, while Acharya and Pedersen (2005) argue that

required returns depend on expected liquidity and liquidity risk. However, cryptocurrencies differ from less frequently traded alternative assets. In private equity, direct real estate and some hedge funds, illiquidity is often linked to lock-ups, slow sale processes, redemption restrictions or high transaction costs. In my cryptocurrency results, the issue appears more connected to volatility, price impact, stablecoin measurement problems and daily-data limitations. Therefore, the evidence suggests a possible cryptocurrency illiquidity premium, but it should be interpreted cautiously.

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